

Evaluation Report - Sprint 1 Week 2

Evaluation Report – Recommender System

1. Introduction

This report evaluates the performance of a recommender system built using user interaction data.

The system generates product recommendations using:

- Popularity-Based Recommendation
- Collaborative Filtering (User-Based)
- Matrix Factorization using SVD

The goal is to compare models and measure effectiveness.

2. Methodology

2.1 Data Preparation

The dataset consists of:

- products_clean.parquet
- user_behavior_clean.parquet

User interactions mapped as:

View = 1

Cart = 5

Purchase = 10

Only users with >5 interactions considered.

2.2 Train-Test Split

Training: 80%

Test: 20%

Example:

Train size: 23982

Test size: 5996

2.3 Evaluation Metrics

Precision@10

Recall@10

MAP@10

3. Results

Popularity:

Precision@10: 0.00643

Recall@10: 0.01873

MAP@10: 0.00606

Collaborative Filtering:

Precision@10: 0.00627

Recall@10: 0.01974

MAP@10: 0.00495

SVD Model:

Precision@10: 0.00606

Recall@10: 0.02095

MAP@10: 0.00670

4. Observations

- Popularity has highest precision
- SVD performs best in recall and MAP
- CF limited due to sparse data
- Metrics low due to implicit feedback
- Trade-off between precision and recall

5. Sample Recommendations

Collaborative Filtering:

Doctor Bed

Compare Bed

Election Wardrobe

Occur Table

SVD:

Doctor Bed

Family Bed

World Chair

Option Wardrobe

Popularity:

Detail Wardrobe

Magazine Sofa

Enjoy Table

Nature Wardrobe

6. Conclusion

- SVD gives best overall performance
- Popularity strong for global trends
- CF limited by sparsity

System successfully demonstrates a full pipeline.

7. Future Improvements

- Time-based splitting
- Hybrid models
- Better feature engineering
- Real-time API deployment